

BRAC University
CSE341 : Microprocessors
Quiz#4

Duration : 25 Minutes

Total Marks : 20

ID:

Name:

Sec:

If XXXX are the first 4 digits of your student ID, consider the following instructions:

- i. MOV AX, [XXXX_h]
- ii. IN AL, [XXXX_h]
- iii. IN AL, DX where DX = XXXX_h
 - a. Find the instructions which are valid and invalid from the given ones, provide your reasoning. [3]
 - b. If instruction (i) and instruction (iii) both are being used for I/O, what are their addressing types? What are their differences? [2 + 3]
 - c. Briefly explain the data transfer process between memory and I/O when 8237 DMAC is being used. [7]
 - d. Suppose, we are using an intel 8237 DMAC which has a faulty DREQ pin. Now we cascade the DMAC with full first-level cascading. How many I/O channels can we use? If the last digit of your student ID is X, consider the X-th I/O channel is also not working. Again, we cascade the second level of DMACs fully. How many I/O channels do we have now? Show your working. [2 + 3]